

SPECTRAL ANALYSIS OF TRANSFER OPERATORS FOR SMOOTH HYPERBOLIC MAPS

CARLANGELO LIVERANI

ABSTRACT. An introduction to the statistical properties of hyperbolic dynamical systems.

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1. INTRODUCTION

In this note we present the so called *functional approach* to the study of the statistical properties of hyperbolic dynamical systems. This is a point of view has its foundation in the study of the Koopman operator (starting from Von Neuman ergodic theorem), further developed in the study of the transfer operator in symbolic dynamics by Sinai, Ruelle and Bowen. A new point of view has emerged with the work of Lasota-Yorke, Ruelle, Keller and, more recently, Kitaev, just to mention a few. This has eventually lead to a substantial revolution in the way of approaching the problems, starting with the paper [BKL].

I present the above range of ideas starting with the case of smooth maps. This will allow to illustrate in the simplest possible setting the power of the transfer operator approach and the type of results that can be obtained once such a machinery is in place. I conclude by discussing how to apply such ideas to the hyperbolic case.

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2. SMOOTH EXPANDING MAPS

By smooth expanding map we mean a map $f \in \mathcal{C}^r(\mathbb{T}, \mathbb{T})$, $r \geq 2$, such that $\inf_x |f'(x)| \geq \lambda > 1$. Clearly $(f, \mathbb{T})$ is a topological dynamical system. Our first goal is to view it as a measurable dynamical system, hence we need to select an invariant probability measure.

Deterministic system often have a lot of invariant measures. In particular, to any periodic orbit is associated an invariant measure. Given such plentiful possibilities, we need a criteria to select the invariant measures that is relevant. A common choice is to consider measures that can be obtained by pushing forward a measure absolutely continuous with respect to Lebesgue.

More preciswely, let $d\mu = h(x)dx$, $h \in L^1(\mathbb{T}^1, \text{Leb})$ and define

$$\begin{aligned}\mu(\varphi) &= \int_{\mathbb{T}} \varphi(x) \mu(dx) \\ f_*\mu(\varphi) &= \mu(\varphi \circ f).\end{aligned}$$

Note that if μ is a probability measure (i.e., $h \geq 0$ and $\mu(1) = 1$), then also $f_*\mu$ is a probability measure. Then

$$\frac{1}{n} \sum_{k=0}^{n-1} f_*^k \mu$$

is a weakly compact set, hence it has accumulation points. One can easily see that such accumulation points are invariant measures for f , that is fixed points for f_* (this is, essentially Krylov-Bogoliubov Theorem). We would then like to study such fixed points.

A simple change of variables shows that $\frac{d(f_*\mu)}{d\text{Leb}} = \mathcal{L}h$ where

$$\mathcal{L}h(x) = \sum_{f(y)=x} \frac{h(y)}{f'(y)}.$$

The operator $\mathcal{L}$ is called the (Ruelle) transfer operator. Of course, to properly define such an operator we must specify on which space it acts. Since

$$\int |\mathcal{L}h(x)| dx \leq \int \mathcal{L}|h|(x) dx = \int |h(x)| dx$$

it follows that $\mathcal{L}$ is well defined as an operator from $L^1(\mathbb{T}, \text{Leb})$ to itself, moreover it is a contraction on $L^1(\mathbb{T}, \text{Leb})$. However, the spectrum of $\mathcal{L}$ on L^1 turns out to be the full unit disk, not a very useful fact.

Following Lasota-Yorke, we look then at the action of $\mathcal{L}$ on $W^{1,1}$.

$$(2.1) \quad \frac{d}{dx} \mathcal{L}h = \mathcal{L} \left(\frac{h}{f'} \right) - \mathcal{L} \left(h \frac{f''}{(f')^2} \right).$$

The above implies the so called *Lasota-Yorke inequalities*

$$(2.2) \quad \begin{aligned}\|\mathcal{L}h\|_{L^1} &\leq \|h\|_{L^1} \\ \|(\mathcal{L}h)'\|_{L^1} &\leq \lambda^{-1} \|h'\|_{L^1} + D \|h\|_{L^1}.\end{aligned}$$

Such inequalities imply that $\mathcal{L}$ is well defined as an operator from $W^{1,1}$ to itself. In addition, when acting on $W^{1,1}$ it is a quasi-compact operator (see Theorem 2.1 for details). That is, the spectrum $\sigma_{W^{1,1}}(\mathcal{L}) \subset \{z \in \mathbb{C} : |z| \leq 1\}$ while the essential spectrum is smaller: $\text{ess-}\sigma_{W^{1,1}}(\mathcal{L}) \subset \{z \in \mathbb{C} : |z| \leq \lambda^{-1}\}$.

To illustrate the above facts, let us consider the simple case in which $D = \|\frac{f''}{(f')^2}\|_{L^\infty}$ is small, more precisely $\lambda^{-1} + D < 1$.

Note that, if $\text{Leb}(h) = 0$, then $\text{Leb}(\mathcal{L}h) = 0$, hence the space $\mathbb{V} = \{h \in L^1 : \text{Leb}(h) = 0\}$ is invariant under $\mathcal{L}$. Also, if $h \in \mathbb{V}$, then, by the mean value theorem, there must exist x_* such that $h(x_*) = 0$, thus

$$\|h\|_{L^1} = \int_{\mathbb{T}} |h(x)| = \int_{\mathbb{T}} \int_{x_*}^x |h'(y)| \leq \|h'\|_{L^1}.$$

Next, let us define the norm $\|h\|_{W^{1,1}} = \|h'\|_{L^1} + a\|h\|_{L^1}$ for some $a > 0$ to be chosen shortly.¹ Then, for $h \in \mathbb{V}$, and $\alpha \in [0, 1]$,

$$\|\mathcal{L}h\|_{W^{1,1}} \leq \lambda^{-1}\|h'\|_{L^1} + (D+a)\|h\|_{L^1} \leq (\lambda^{-1}+D+a)\|h'\|_{L^1} \leq (\lambda^{-1}+D+a)\|h\|_{W^{1,1}}.$$

We can then choose a such that $\lambda^{-1} + D + a < 1$, and we have that $\mathcal{L}$ is a contraction on $\mathbb{V}$. Since $\mathcal{L}'\text{Leb} = \text{Leb}$, hence $1 \in \sigma(\mathcal{L})$, we have that there exists $h_* \in L^1$ such that $\mathcal{L}h = h_*\text{Leb}(h) + Qh$, where $\|Q\|_{W^{1,1}} < 1$ and $\text{Leb}Q = Qh_* = 0$. We have just proven that $h_*(x)dx$ is the only invariant measure of f absolutely continuous with respect to Lebesgue.²

In fact, the above spectral decomposition, and hence the uniqueness of the invariant measure absolutely continuous with respect to Lebesgue, holds in much higher generality, in particular for each $f \in \mathcal{C}^2$ such that $|f'| \geq \lambda > 1$ due to the following theorem.

Theorem 2.1 ([?]). *Let $\mathcal{B} \subset \mathcal{B}_w$ two Banach spaces, let $\|\cdot\|$ and $\|\cdot\|_w$ be the respective norms, such that $\{v \in \mathcal{B} : \|v\| \leq 1\}$ is relatively compact in $\mathcal{B}_w$. In addition, let $\mathcal{L} : \mathcal{B} \rightarrow \mathcal{B}$ be a linear operator such that, for each $n \in \mathbb{N}$ and $v \in \mathcal{B}$,*

$$\begin{aligned} \|\mathcal{L}^n v\|_w &\leq C_{\#}\|v\|_w \\ \|\mathcal{L}^n v\| &\leq C_{\#}\lambda^{-n}\|v\| + B\|v\|_w \end{aligned}$$

then $\mathcal{L}$ is a quasi compact operator with essential spectral radius bounded by λ^{-1} .

Since it is not hard to show that smooth expanding maps are mixing, it follows that $\mathcal{L}$ cannot have eigenvalues of modulus one and hence must mix exponentially fast (see [1] for an exhaustive discussion).

Problem 2.2. *Derive further (2.1) to obtain a Lasota-Yorke inequality with respect to the norms $W^{p,1}$, $W^{p-1,1}$, $p \leq r-1$. Show then that the essential spectral radius of $\mathcal{L}$ when acting on $W^{p,1}$ is bounded by λ^{-p} .*

2.1. The Central Limit Theorem. We can now address the second question we have posed. From the above discussion is clear that we must chose $c_n = \sqrt{n}$.

Let $f \in W^{1,1}$ and set $\hat{f} := f - \mu(f)$, then

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \hat{f} \circ T^k(x) = 0 \quad m - \text{a.e.}$$

Let us set $\Psi_n := \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \hat{f} \circ T^k$. We can consider Ψ_n a random variable with distribution $F_n(t) := \mu(\{x : \Psi_n(x) \leq t\})$. It is well know that, for each

¹ Note that all such norms are equivalent, so the choice of a special value of a is only a matter of convenience.

² To make the argument precise use that $W^{1,1}$ is dense in L^1 .

continuous function g holds³

$$\mu(g(\Psi_n)) = \int_{\mathbb{R}} g(t) dF_n(t)$$

where the integral is a Riemann-Stieltjes integral. It is thus clear that if we can control the distribution F_n , we have a very sharp understanding of the probability to have small deviations (of order $\sqrt{n}$) from the limit. From the work in the previous section it follows that there exists $\delta > 0$ such that, for each $|\lambda| \leq \delta\sqrt{n}$,

$$(2.3) \quad \begin{aligned} \varphi_n(\lambda) &:= \mu(e^{i\lambda\Psi_n}) = \mu(\mathcal{L}_{i\lambda/\sqrt{n}}^n h) = (1 - \frac{\sigma^2\lambda^2}{2n} + \mathcal{O}(\lambda^3 n^{-\frac{3}{2}} + \rho^n)) \|f\|_{W^{1,1}}^n \\ &= e^{-\frac{\sigma^2\lambda^2}{2}} (1 + \mathcal{O}(\lambda^3 n^{-\frac{1}{2}} + n\rho^n)) \|f\|_{W^{1,1}}. \end{aligned}$$

The above quantity is called *characteristic function* of the random variable and determine the distribution via the formula

$$F_n(b) - F_n(a) = \lim_{\Lambda \rightarrow \infty} \frac{1}{2\pi} \int_{-\Lambda}^{\Lambda} \frac{e^{-ia\lambda} - e^{-ib\lambda}}{i\lambda} \varphi_n(\lambda) d\lambda,$$

as can be seen in any basic book of probability theory.⁴

Formula (2.3) means in particular that

$$\lim_{n \rightarrow \infty} m(e^{\lambda\Psi_n}) = e^{-\frac{\sigma^2\lambda^2}{2}} =: \varphi(\lambda).$$

What can we infer out of the above facts? First of all a simple computation shows that

$$g(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-it\lambda} \varphi(\lambda) d\lambda = \frac{1}{\sqrt{\pi}\sigma} e^{-\frac{t^2}{2\sigma^2}}$$

a random variable with such a density is called a Gaussian random variable with zero average and variance σ . Accordingly, formula (2.3) can be interpreted by saying that there exists a Gaussian random variable G such that

$$\frac{1}{n} \sum_{k=0}^{n-1} \hat{f} \circ T^k \sim \frac{1}{\sqrt{n}} G(1 + \mathcal{O}(n^{-\frac{1}{2}}))$$

in distribution. But what does this means concretely. Actual estimates are made difficult by the fact that the distribution under study no not necessarily have a density, thus we are Fourier transforming function that behave quite badly at infinity. To overcome such a problem we can smoothen the quantities involved.

³If $g \in C_0^1$, then

$$\int_{\mathbb{R}} g dF_n = - \int_{\mathbb{R}} F_n(t) g'(t) dt = - \int_{\mathbb{R}} dt \int_{\mathbb{T}^1} dx \chi_{\{z : \Psi_n(z) \leq t\}}(x) g'(t).$$

Applying Fubini yields

$$\int_{\mathbb{R}} g dF_n = - \int_{\mathbb{T}^1} dx \int_{\mathbb{R}} dt \chi_{\{z : \Psi_n(z) \leq t\}}(x) g'(t) = - \int_{\mathbb{T}^1} dx \int_{\Psi_n(x)}^{\infty} g'(t) dt = \int_{\mathbb{T}^1} dx g(\Psi_n(x)).$$

⁴In the case when there exists a density, that is an L^1 function f_n such that $F_n(b) - F_n(a) = \int_a^b f_n(t) dt$, then the formula above becomes simply

$$f_n(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-it\lambda} \varphi_n(\lambda) d\lambda,$$

and follows trivially by the inversion of the Fourier transform.

Let $j \in \mathcal{C}^\infty(\mathbb{R}, \mathbb{R}_+)$ such that $\int_{\mathbb{R}} j(t) dt = 1$, $j(t) = j(-t)$, and $j(t) = 0$ for all $|t| > 1$, for each $\varepsilon > 0$ defined then $j_\varepsilon(t) := \varepsilon^{-1} j(\varepsilon^{-1} t)$ and

$$(2.4) \quad F_{n,\varepsilon}(t) := \int_{\mathbb{R}} j_\varepsilon(t-s) F_n(s) ds.$$

A simple computation shows that, for each $a, b \in \mathbb{R}$, holds

$$F_n(b+\varepsilon) - F_n(a-\varepsilon) \geq F_{n,\varepsilon}(b) - F_{n,\varepsilon}(a) \geq F_n(b-\varepsilon) - F_n(a+\varepsilon)$$

that is: if the measurements have a precision worst than 2ε , then $F_{n,\varepsilon}$ is as good as F_n to describe the resulting statistics. On the other hand calling $\varphi_{n,\varepsilon}$ the characteristic function associated to $F_{n,\varepsilon}$, holds $\varphi_{n,\varepsilon}(\lambda) = \varphi_n(\lambda) \hat{j}(\varepsilon\lambda)$, where $\hat{j}$ is the Fourier transform of j . Since now $F_{n,\varepsilon}$ is the law of a smooth random variable it has a density $f_{n,\varepsilon}$ and

$$f_{n,\varepsilon}(t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\lambda t} \varphi_n(\lambda) \hat{j}(\varepsilon\lambda) d\lambda$$

since j is smooth it follows that there exists $C > 0$ such that $|\hat{j}(\lambda)| \leq C(1+\lambda^2)^{-2}$. We can finally use formula (2.3) to obtain a quantitative estimate

$$\begin{aligned} f_{n,\varepsilon}(t) &= \frac{1}{2\pi} \int_{-\varepsilon\sqrt{n}}^{\varepsilon\sqrt{n}} e^{-i\lambda t} \varphi_n(\lambda) \hat{j}(\varepsilon\lambda) d\lambda + \mathcal{O}(\varepsilon^{-5} n^{-\frac{3}{2}}) \\ &= \frac{1}{2\pi} \int_{-\varepsilon\sqrt{n}}^{\varepsilon\sqrt{n}} e^{-i\lambda t} \varphi(\lambda) \hat{j}(\varepsilon\lambda) d\lambda + \mathcal{O}(\varepsilon^{-5} n^{-\frac{3}{2}} + n^{-\frac{1}{2}}) \\ &= g(t) + \mathcal{O}(\varepsilon + \varepsilon^{-5} n^{-\frac{3}{2}} + n^{-\frac{1}{2}}) = g(t) + \mathcal{O}(n^{-\frac{1}{2}}) \end{aligned}$$

provided we choose $n^{-\frac{1}{2}} \geq \varepsilon \geq n^{-5}$. Which, as announced, means that, if the precision of the instrument is compatible with the statistics, the typical fluctuations in measurements are of order $\frac{1}{\sqrt{n}}$ and Gaussian. This is well known by sperimentalists who routinely assume that the result of a measurement is distributed according to a Gaussian.⁵

2.2. Perturbation theory. To answer the questions posed at the beginning we need some perturbation theorems. Few such results are available (e.g., see [18], [26] for a review and [6] for some more recent results), here we will follow mainly the theory developed in [17] adapted to the special cases at hand.

For simplicity let us work directly with the densities and in the case $d = 1$. Then $\mathcal{L}$ is the transfer operator for the densities. We will start by considering an abstract family of operators $\mathcal{L}_\varepsilon$ satisfying the following properties.

Condition 1. *Consider a family of operators $\mathcal{L}_\varepsilon$ with the following properties*

(1) *A uniform Lasota-Yorke inequality:*

$$\|\mathcal{L}_\varepsilon^n h\|_{W^{1,1}} \leq A\lambda^{-n} \|h\|_{W^{1,1}} + B|h|_{L^1}, \quad |\mathcal{L}_\varepsilon^n h|_{L^1} \leq C|h|_{L^1};$$

(2) $\int \mathcal{L}h(x) dx = \int h(x) dx$;

⁵Note however that our proof holds in a very special case that has little to do with a real experimental setting. To prove the analogous statement in for a realistic experiment is a completely different ball game.

(3) For $L : W^{1,1} \rightarrow W^{1,1}$ define the norm

$$|||L||| := \sup_{\|h\|_{W^{1,1}} \leq 1} |Lf|_{L^1},$$

that is the norm of L as an operator from $W^{1,1} \rightarrow L^1$. Then we require that there exists $D > 0$ such that

$$|||\mathcal{L} - \mathcal{L}_\varepsilon||| \leq D\varepsilon.$$

Condition 1-(3) specifies in which sense the family $\mathcal{L}_\varepsilon$ can be considered an approximation of the unperturbed operator $\mathcal{L}$. Notice that the condition is rather weak, in particular the distance between $\mathcal{L}_\varepsilon$ and $\mathcal{L}$ as operators on $W^{1,1}$ can be always larger than 1. Such a notion of closeness is completely inadequate to apply standard perturbation theory, to get some perturbations results it is then necessary to drastically restrict the type of perturbations allowed, this is done by Conditions 1-(1,2) which state that all the approximating operators enjoys properties very similar to the limiting one.⁶

To state a precise result consider, for each operator L , the set

$$V_{\delta,r}(L) := \{z \in \mathbb{C} \mid |z| \leq r \text{ or } \text{dist}(z, \sigma(L)) \leq \delta\}.$$

Since the complement of $V_{\delta,r}(L)$ belongs to the resolvent of L it follows that

$$H_{\delta,r}(L) := \sup \{ \|(z - L)^{-1}\|_{W^{1,1}} \mid z \in \mathbb{C} \setminus V_{\delta,r}(L) \} < \infty.$$

By $R(z)$ and $R_\varepsilon(z)$ we will mean respectively $(z - \mathcal{L})^{-1}$ and $(z - \mathcal{L}_\varepsilon)^{-1}$.

Theorem 2.3 ([17]). *Consider a family of operators $\mathcal{L}_\varepsilon : W^{1,1} \rightarrow W^{1,1}$ satisfying Conditions 1. Let $H_{\delta,r} := H_{\delta,r}(\mathcal{L})$; $V_{\delta,r} := V_{\delta,r}(\mathcal{L})$, $r > \lambda^{-1}$, $\delta > 0$, then, if $\varepsilon \leq \varepsilon_1(\mathcal{L}, r, \delta)$, $\sigma(\mathcal{L}_\varepsilon) \subset V_{\delta,r}(\mathcal{L})$. In addition, if $\varepsilon \leq \varepsilon_0(\mathcal{L}, r, \delta)$, there exists a $a > 0$ such that, for each $z \notin V_{\delta,r}$, holds true*

$$|||R(z) - R_\varepsilon(z)||| \leq C\varepsilon^a.$$

*Proof.*⁷ To start with we collect some trivial, but very useful algebraic identities.

For each operator $L : W^{1,1} \rightarrow W^{1,1}$ and $n \in \mathbb{Z}$ holds

$$(2.5) \quad \frac{1}{z} \sum_{i=0}^{n-1} (z^{-1}L)^i (z - L) + (z^{-1}L)^n = \mathbf{1}$$

$$(2.6) \quad R(z)(z - \mathcal{L}_\varepsilon) + \frac{1}{z} \sum_{i=0}^{n-1} (z^{-1}\mathcal{L})^i (\mathcal{L}_\varepsilon - \mathcal{L}) + R(z)(z^{-1}\mathcal{L})^n (\mathcal{L}_\varepsilon - \mathcal{L}) = \mathbf{1}$$

$$(2.7) \quad (z - \mathcal{L}_\varepsilon) [G_{n,\varepsilon} + (z^{-1}\mathcal{L}_\varepsilon)^n R(z)] = \mathbf{1} - (z^{-1}\mathcal{L}_\varepsilon)^n (\mathcal{L}_\varepsilon - \mathcal{L}) R(z)$$

$$(2.8) \quad [G_{n,\varepsilon} + (z^{-1}\mathcal{L}_\varepsilon)^n R(z)] (z - \mathcal{L}_\varepsilon) = \mathbf{1} - (z^{-1}\mathcal{L}_\varepsilon)^n R(z) (\mathcal{L}_\varepsilon - \mathcal{L}),$$

where we have set $G_{n,\varepsilon} := \frac{1}{z} \sum_{i=0}^{n-1} (z^{-1}\mathcal{L}_\varepsilon)^i$.

⁶Actually only Condition 1-(1) is needed in the following. Condition 1-(2) simply implies that the eigenvalue one is common to all the operators. If 1-(2) is not assumed, then the operator $\mathcal{L}_\varepsilon$ will always have one eigenvalue close to one, but the spectral radius could vary slightly, see [22] for such a situation.

⁷This proof is simpler than the one in [17], yet it gives worst bounds, although sufficient for the present purposes.

Let us start applying the above formulae. For each $h \in W^{1,1}$ and $z \notin V_{r,\delta}$ holds

$$\begin{aligned} \|(z^{-1}\mathcal{L}_\varepsilon)^n(\mathcal{L}_\varepsilon - \mathcal{L})R(z)h\|_{W^{1,1}} &\leq (r\lambda)^{-n}A\|(\mathcal{L}_\varepsilon - \mathcal{L})R(z)h\|_{W^{1,1}} + \frac{B}{r^n}|(\mathcal{L}_\varepsilon - \mathcal{L})R(z)h|_{L^1} \\ &\leq [(r\lambda)^{-n}A2C_1 + Br^{-n}D\varepsilon]H_{r,\delta}\|h\|_{W^{1,1}} < \|h\|_{W^{1,1}} \end{aligned}$$

Thus $\|(z^{-1}\mathcal{L}_\varepsilon)^n(\mathcal{L}_\varepsilon - \mathcal{L})R(z)\|_{W^{1,1}} < 1$ and the operator on the right hand side of (2.7) can be inverted by the usual Neumann series. Accordingly, $(z - \mathcal{L}_\varepsilon)$ has a well defined right inverse. Analogously,

$$\|(z^{-1}\mathcal{L}_\varepsilon)^n R(z)(\mathcal{L}_\varepsilon - \mathcal{L})h\|_{W^{1,1}} \leq (r\lambda)^{-n}A\|R(z)(\mathcal{L}_\varepsilon - \mathcal{L})h\|_{W^{1,1}} + Br^{-n}|R(z)(\mathcal{L}_\varepsilon - \mathcal{L})h|_{L^1}.$$

This time to continue we need some informations on the L^1 norm of the resolvent. Let $g \in W^{1,1}$, then equation (2.5) yields

$$\begin{aligned} |R(z)g|_{L^1} &\leq \frac{1}{r} \sum_{i=0}^{n-1} |(z^{-1}\mathcal{L})^i g|_{L^1} + \|R(z)(z^{-1}\mathcal{L})^n g\|_{W^{1,1}} \\ &\leq \frac{1}{r^n(1-r)}|g|_{L^1} + H_{\delta,r}A(r\lambda)^{-n}\|g\|_{W^{1,1}} + H_{\delta,r}Br^{-n}|g|_{L^1} \\ &\leq r^{-n}(H_{\delta,r}B + (1-r)^{-1})|g|_{L^1} + H_{\delta,r}A(r\lambda)^{-n}\|g\|_{W^{1,1}} \end{aligned}$$

Substituting, we have

$$\begin{aligned} \|(z^{-1}\mathcal{L}_\varepsilon)^n R(z)(\mathcal{L}_\varepsilon - \mathcal{L})h\|_{W^{1,1}} &\leq \{(r\lambda)^{-n}AH_{\delta,r}2C_1[1 + Br^{-n}] \\ &\quad + Br^{-2n}[H_{\delta,r}B + (1-r)^{-1}]D\varepsilon\}\|h\|_{W^{1,1}} < 1, \end{aligned}$$

again, provided ε is small enough and choosing n appropriately. Hence the operator on the right hand side of (2.8) can be inverted, thereby providing a left inverse for $(z - \mathcal{L}_\varepsilon)$.⁸ This implies that z does not belong to the spectrum of $\mathcal{L}_\varepsilon$.

To investigate the second statement note that (2.6) implies

$$R(z) - R_\varepsilon(z) = \frac{1}{z} \sum_{i=0}^{n-1} (z^{-1}\mathcal{L})^i (\mathcal{L}_\varepsilon - \mathcal{L})R_\varepsilon(z) - R(z)(z^{-1}\mathcal{L})^n (\mathcal{L}_\varepsilon - \mathcal{L})R_\varepsilon(z).$$

Accordingly, for each $\varphi \in W^{1,1}$ holds

$$|R(z)\varphi - R_\varepsilon(z)\varphi|_{L^1} \leq \{r^{-n}(1-r)^{-1}\varepsilon + H_{\delta,r}(\lambda r)^{-n}2AC_1 + H_{\delta,r}B\varepsilon\}\|R_\varepsilon(z)\varphi\|_{W^{1,1}}.$$

□

2.3. Deterministic stability. The $\mathcal{L}_\varepsilon$ are Perron-Frobenius (Transfer) operators of maps T_ε which are $\mathcal{C}^1$ -close to T , that is $d_{\mathcal{C}^1}(T_\varepsilon, T) = \varepsilon$ and such that $d_{\mathcal{C}^2}(T_\varepsilon, T) \leq M$, for some fixed $M > 0$. In this case the uniform Lasota-Yorke inequality is trivial. On the other hand, for all $\varphi \in \mathcal{C}^1$ holds

$$\int (\mathcal{L}_\varepsilon f - \mathcal{L}f)\varphi = \int f(\varphi \circ T_\varepsilon - \varphi \circ T).$$

Now let $\Phi(x) := (D_x T)^{-1} \int_{T_x}^{T_\varepsilon x} \varphi(z) dz$, since

$$\Phi'(x) = -(D_x T)^{-1} D_x^2 T \Phi(x) + D_x T_\varepsilon (D_x T)^{-1} \varphi(T_\varepsilon x) - \varphi(Tx)$$

follows

$$\int (\mathcal{L}_\varepsilon f - \mathcal{L}f)\varphi = \int f\Phi' + \int f(x)[(D_x T)^{-1} D_x^2 T \Phi(x) + (1 - D_x T_\varepsilon (D_x T)^{-1})\varphi(T_\varepsilon x)].$$

⁸ Note that, for any operator A , if R, L are such that $LA = AR = \mathbf{1}$. Then $R = LAR = L$, hence A is invertible.

Given that $|\Phi|_\infty \leq \lambda^{-1}\varepsilon|\varphi|_\infty$ and $|1 - D_x T_\varepsilon(D_x T)^{-1}|_\infty \leq \lambda^{-1}\varepsilon$, we have

$$\int (\mathcal{L}_\varepsilon f - \mathcal{L}f)\varphi \leq \|f\|_{W^{1,1}}\lambda^{-1}|\varphi|_\infty\varepsilon + \|f\|_{L^1}\lambda^{-1}(B+1)\varepsilon|\varphi|_\infty \leq D\|f\|_{W^{1,1}}\varepsilon|\varphi|_\infty.$$

By Lebesgue dominate convergence theorem we obtain the above inequality for each $\varphi \in L^\infty$, and taking the sup on such φ yields the wanted inequality.

$$|\mathcal{L}_\varepsilon f - \mathcal{L}f|_{L^1} \leq D\|f\|_{W^{1,1}}\varepsilon.$$

We have thus seen that all the requirements in Condition 1 are satisfied. See [16] for a more general setting including piecewise smooth maps.

2.4. Stochastic stability. Next consider a set of maps $\{T_\omega\}$ depending on a parameter $\omega \in \Omega$. In addition assume that Ω is a probability space and consider a measure P on Ω . Consider the process $x_n = T_{\omega_n} \circ \dots \circ T_{\omega_1} x_0$ where the ω are i.i.d. random variables distributed accordingly to P and let E_μ be the expectation of such process when x_0 is distributed according to μ . Then, calling $\mathcal{L}_\omega$ the transfer operator associated to T_ω , we have

$$E(f(x_{n+1}) | x_n) = \mathcal{L}_P f(x_n) := \int_\Omega \mathcal{L}_\omega f(x_n) P(d\omega).$$

Then if

$$|\mathcal{L}_\omega h|_{W^{1,1}} \leq \lambda_\omega^{-1}|h|_{W^{1,1}} + B_\omega|h|_{L^1}$$

integrating yields

$$|\mathcal{L}_P h|_{W^{1,1}} \leq E(\lambda_\omega^{-1})|h|_{W^{1,1}} + E(B_\omega)|h|_{L^1}$$

And the operator $\mathcal{L}_P$ satisfy a Lasota-Yorke inequality provided that $E(\lambda^{-1}) < 1$ and $E(B) < \infty$.

In addition, if for some map T and associated transfer operator $\mathcal{L}$,

$$E(|\mathcal{L}_\omega h - \mathcal{L}h|) \leq \varepsilon|h|_{W^{1,1}}$$

then we can apply perturbation theory and obtain stochastic stability.

2.5. Computability. If we want to compute the invariant measure and the rate of decay of correlations, we can use the operator P_t defined in (??) and define $\mathcal{L}_{t,m} = P_t \mathcal{L}^m$. By the estimates in Lemma ?? it follows

$$|\mathcal{L}_{t,m} h|_{W^{1,1}} \leq 4^d \sigma^m |h|_{W^{1,1}} + B|h|_{L^1}.$$

We can then chose the smallest m so that $4^d \sigma^m = \sigma_1 < 1$. Moreover, we also saw that

$$|\mathcal{L}_{t,m} h - \mathcal{L}h| \leq t^{-1}|h|_{W^{1,1}}.$$

So we are again in the realm of our perturbation theory and we have that the finite dimensional operator $\mathcal{L}_{t,m}$ has spectrum close to the one of the transfer operator. We can then obtain all the info we want by diagonalizing a matrix.

2.6. Linear response. Linear response is a theory widely used by physicists. In essence it says the follow: consider a one parameter family of systems T_s and the associated (e.g.) invariant measures μ_s , then, for a given observable f one want to study the response of the system to a small change in s , and, not surprisingly, one expects $\mu_s(f) = \mu_0(f) + s\nu(f) + o(s)$. That is one expects differentiability in s . Yet differentiability is not ensured by Theorem 2.3. Is it possible to ensure conditions under which linear response holds? The answer is yes (for example if holds if the maps are sufficiently smooth and the dependence on the parameter is also smooth in an appropriate sense). To prove it one need a sophistication of Theorem 2.3 that can be found in [20].

3. THE CONTRACTING CASE

Having illustrated the power of the transfer operator approach in the expanding case, it is natural to investigate to which extent it can be generalised. A first remark is that, when it works, it automatically implies that the system either does not mix or mixes exponentially fast. Accordingly, the direct application of the above strategy is ill suited for cases where the decay of correlation is only polynomial (although one can apply it after inducing).

On the contrary, when the decay of correlations is expected to be exponential, one can reasonably try to implement a transfer operator approach directly. In particular, it is natural to investigate the possibility to apply it to uniformly hyperbolic systems and partially hyperbolic system. To this end there are several technical difficulties, some of them still outstanding.

Clearly the initial obstacle is posed by the existence of contracting directions. So the first question is: can we find appropriate Banach spaces for which the transfer operator of a contracting map has good spectral properties? The answer is yes. In fact, again, there are several possibilities.⁹ Let us illustrate a basic one in the simplest possible case: let $f \in \mathcal{C}^3(\mathbb{T}, \mathbb{T})$ be a orientation preserving diffeomorphism with two fixed points, one attracting and one repelling. Without loss of generality we can assume that zero is the attracting fixed point. Let $\psi \in \mathcal{C}^2(\mathbb{T}, \mathbb{R})$ be a positive function such that $\psi = 1$ in a neighborhood of zero and $\psi = 0$ in a neighborhood of the repelling fixed point. Also let us assume that the support of ψ be small enough so that

$$\|\psi f'\|_{\mathcal{C}^0} \leq \lambda^{-1} < 1.$$

Consider the transfer operator $\mathcal{L}h = (\psi h[f']^{-1}) \circ f^{-1}$. Note that $\mathcal{L}\delta_0 = \delta_0$, thus the spectral radius on $\mathcal{C}^0(\mathbb{T}, \mathbb{R})'$ is at least one. However, as in the previous example, to obtain a Lasota-Yorke inequality we need to consider the operator acting on a different space. This time the space cannot be $\mathcal{C}^1$ otherwise we would obtain a spectral radius larger than one. We need an idea.

The idea is to let it act on $(\mathcal{C}^1)'$, the dual of $\mathcal{C}^1$. Then, for each $\varphi \in \mathcal{C}^1$, $\|\varphi\|_{\mathcal{C}^1} \leq 1$, we can write

$$\int \varphi \mathcal{L}h = \int \varphi \circ f \psi h.$$

Note that $\|\varphi \circ f \psi\|_{\mathcal{C}^0} \leq \|\varphi\|_{\mathcal{C}^0}$ while $\|(\varphi \circ f \psi)'\|_{\mathcal{C}^0} \leq \lambda^{-1} \|\varphi'\|_{\mathcal{C}^0} + C_{\#} \|\varphi\|_{\mathcal{C}^0}$. The above gives a promising estimate for the derivative but, to obtain a contraction, it is necessary to reduce the $\mathcal{C}^0$ norm as well. To this end note that, for each $\varepsilon > 0$

⁹ To be honest they all have the same flavour, although they might be quite different in the details.

there exists $\varphi_\varepsilon \in \mathcal{C}^2$ such that $\|\varphi_\varepsilon\|_{\mathcal{C}^1} \leq 1$ and $\|\varphi - \varphi_\varepsilon\|_{\mathcal{C}^0} \leq \varepsilon$.¹⁰ Then, there exists $B_0 > 0$ such that

$$\left| \int \varphi \mathcal{L}h \right| \leq \int |(\varphi - \varphi_\varepsilon) \circ f \psi h| + \left| \int \varphi_\varepsilon \circ f \psi h \right| \leq 2\lambda^{-1} \|h\|_{(\mathcal{C}^1)'} + B_0 \|h\|_{(\mathcal{C}^2)'}$$

where we have chosen ε small enough.

Problem 3.1. *Use computations similar to the above to show that there exists $C, B > 0$ such that, for all $n \in \mathbb{N}$ and $h \in (\mathcal{C}^1)'$,*

$$(3.1) \quad \begin{aligned} \|\mathcal{L}^n h\|_{(\mathcal{C}^2)'} &\leq C \|h\|_{(\mathcal{C}^2)'} \\ \|\mathcal{L}^h\|_{(\mathcal{C}^1)'} &\leq C\lambda^{-n} \|h\|_{(\mathcal{C}^1)'} + B \|h\|_{(\mathcal{C}^2)'} \end{aligned}$$

Problem 3.2. *Show that the unit ball $\{h \in (\mathcal{C}^1)' : \|h\|_{(\mathcal{C}^1)'}\} is relatively compact in $(\mathcal{C}^2)'$.$*

Problems 3.1, 3.2 and Theorem 2.1 imply that $\mathcal{L}$, when acting on $(\mathcal{C}^1)'$, has spectral radius one and essential spectral radius bounded by λ^{-1} . We have already seen that one belongs to the spectra. Suppose that $e^{i\theta}$ is in the spectra, then there exists $h_\theta \in (\mathcal{C}^1)'$ such that, for all $\varphi \in \mathcal{C}^1$ and $n \in \mathbb{N}$,

$$\int e^{i\theta n} h_\theta \varphi = \int \mathcal{L}^n h_\theta \varphi = \int h_\theta \left[\prod_{k=0}^{n-1} \psi \circ f^k \right] \varphi \circ f^n.$$

Note that, if $\text{supp } \varphi \cap \{0\} = \emptyset$, then there exists n large enough so that $\psi \circ \varphi \circ f^n = 0$. This implies that $\text{supp } h_\theta = \{0\}$. In other words the only possibility is $\int h_\theta \varphi = a\varphi(0) + b\varphi'(0)$. But then we must have, for all $\varphi \in \mathcal{C}^1$,

$$e^{i\theta} [a\varphi(0) + b\varphi'(0)] = a\varphi(0) + b\varphi'(0)f'(0)$$

which has a solution only for $\theta = 0$ and $b = 0$. In other words one is the only eigenvalue of modulus one and it is a simple eigenvalue. It follows that the system is exponentially mixing.¹¹ Moreover, all the transfer operator theory previous developed can be applied to this situation. Indeed is a good exercise to do so.

4. AN INTERLUDE: TORAL AUTOMORPHISMS

The next step is to treat higher dimensional systems in which both contraction and expansion are present. The simplest such case is called the *hyperbolic* case in which only expanding and contraction directions are present. Before describing some element of the general theory we discuss in detail the simplest possible example: Toral automorphisms. For such simple systems we will discuss three different approaches that illustrate the basis of three different general theories used to investigate the statistical properties of dynamical systems.

Let us consider the map from $\mathbb{T}^2$ to itself defined by

$$f(x) = Ax \mod 1,$$

¹⁰ Simply use a mollifier.

¹¹ To be precise it is exponentially mixing for observables that are supported away from the expanding fixed point. Given the above estimates, it is a simple exercise to study what happens to a general observable.

with $A \in SL(2, \mathbb{Z})$. Also, for simplicity, let us assume that $A^t = A$ and $A_{i,j} > 0$. In analogy with the previous section we can define the operator $\mathcal{L}h = h \circ f^{-1}$, note that

$$\int_{\mathbb{T}^2} \varphi \mathcal{L}h = \int_{\mathbb{T}^2} \varphi \circ f \cdot h.$$

In fact, the reader can consider, as a concrete example,

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Note that the lebesgue measure is invariant since $\det(A) = 1$. Moreover $\text{Tr}(A) > 2$. Accordingly, the characteristic polynomial reads $t^2 + \text{Tr}(A)t + 1$, which has roots λ, λ^{-1} , $\lambda > 1$. We call then v^u, v^s the two normalized vectors such that

$$(4.1) \quad \begin{aligned} Av^u &= \lambda v^u \\ Av^s &= \lambda^{-1} v^s. \end{aligned}$$

Note that, since the matrix is assumed symmetric, $\langle v^u, v^s \rangle = 0$.

We have thus a natural reference measure. In fact, $(f, \mathbb{T}^2, \text{Leb})$ turns out to be *mixing*, that is: for each $h, \varphi \in \mathcal{C}^0$

$$\lim_{n \rightarrow \infty} \int_{\mathbb{T}^2} h(x) \varphi(f^n(x)) dx = \int_{\mathbb{T}^2} h(x) dx \int_{\mathbb{T}^2} \varphi(x) dx.$$

In alternative, the mixing can be stated in the following equivalent way: for each probability measure μ such that $\frac{d\mu}{d\text{Leb}} = h \in L^1$ and, for each $\varphi \in \mathcal{C}^0$,¹²

$$(4.2) \quad \lim_{n \rightarrow \infty} \int_{\mathbb{T}^2} f_*^n \mu(\varphi) = \int_{\mathbb{T}^2} \varphi(x) dx.$$

This is a very relevant property from the applied point of view as it says that asymptotically our system is described by the Lebesgue measure but, of course, this is really useful only if the speed in the converge of the limit is fast enough. Form this consideration follows the basic question that we want to address in the following:

What is the speed of convergence in the limit (4.2) ?

4.1. Standar pairs. The basic idea is to consider a special class of measures that behaves, in some appropriate, similarly to the case of expanding maps. This have been introduced, in the present form, by Dolgopyat and are called *standard pairs*.

Fix some $a > 1$ and define

$$D_a = \left\{ h \in \mathcal{C}^0(\mathbb{R}, \mathbb{R}_+) : \int_{\mathbb{T}^2} h(x) dx = 1; \forall t, s \in \mathbb{R}, \frac{h(t)}{h(s)} \leq e^{a|t-s|} \right\}.$$

Also, for each $b \in \mathbb{R}_+$, $x \in \mathbb{T}^2$ and $h \in \mathcal{C}^0(\mathbb{R}, \mathbb{R}_+)$ define the measure

$$\mu_{b,x,h}(\varphi) = \int_{-b}^b h(t) \varphi(x + tv^u) dt.$$

The the standard pairs are the set

$$S_a = \{ \mu_{b,x,h} : b \in [1, 2], x \in \mathbb{T}^2, h \in D_a \}.$$

¹² Recall that $\mu(\varphi) = \int_{\mathbb{T}^2} \varphi(x) h(x) dx$ and $f_* \mu(\varphi) = \mu(f \circ \varphi)$.

The above are our building blocks, let us see what we can construct with them. First of all we can take the convex hull: for each fine set $\{p_i\}$ of positive numbers such that $\sum_i p_i = 1$ and set $\{\mu_i\} \subset S_a$ we can consider the probability measure

$$\mu = \sum_i p_i \mu_i.$$

Such measures are called *standard families* and we will call CS_a the set of all standard families. The first important fact is the following.

Lemma 4.1. *The Lebesgue measure belongs to the weak closure of CS_a .*¹³

Proof. Note that, setting $v^u = (1 + u^2)^{-\frac{1}{2}}(1, u)$, for each $\varphi \in \mathcal{C}^0$,

$$\text{Leb}(\varphi) = \int_0^1 dt \int_0^1 ds \varphi(t, s + ut) = \int_0^1 ds \int_0^{\sqrt{1+u^2}} dt \varphi(se_2 + tv^u).$$

Note that the second integral can be written as the convex combination of finitely many standard pairs, the results follows then since the integral is the limit of finite sums. $\square$

Next we want to know how the standard pairs behaves under push forward.

Lemma 4.2. *For each $n \in \mathbb{N}$ and $\mu \in CS_a$ it holds true $f_*^n \mu \in CS_{\lambda^{-n}a}$.*

Proof. Note that is suffice to prove that if $\mu \in S_a$ then $f_*^n \mu \in CS_a$. Then, recalling (4.1),

$$f_*^n \mu_{b,x,h}(\varphi) = \int_{-b}^b h(t) \varphi(f(x) + t\lambda^n v^u) = \int_{-\lambda^n b}^{\lambda^n b} h(t\lambda^{-n}) \varphi(f(x) + tv^u).$$

Next, let $c \in [1, 2]$ and $K \in \mathbb{N}$ such that $\lambda^n b = Kc$ and define $t_i = -\lambda^n b + (2i+1)c$. We can then write

$$\begin{aligned} f_*^n \mu_{b,x,h}(\varphi) &= \sum_{i=0}^{K-1} p_i \int_{-c}^c h_i(t) \varphi([f(x) + t_i v^u] + tv^u) dt \\ p_i &= \int_{-c}^c h(\lambda^{-n}(t_i + t)) dt \\ h_i(t) &= p_i^{-1} h(\lambda^{-n}(t_i + t)). \end{aligned}$$

Accordingly, the Lemma is proven provided $h_i \in D_{\lambda^{-n}a}$. This follows from

$$\frac{h_i(t)}{h_i(s)} = \frac{h(\lambda^{-n}(t_i + t))}{h(\lambda^{-n}(t_i + s))} \leq e^{a\lambda^{-n}|t-s|}.$$

$\square$

Remark 4.3. *Note that the unbounded contraction in n proven in the previous Lemma is a peculiarity of the linear systems we are studying, however in the non-linear case a fixed contraction takes place and this is all we will use in the following.*

Next we call two standard pairs $\mu_1 = \mu_{b,x,h}$, $\mu_2 = \mu_{b,x+sv^s,h}$, $s \in [1, 2]$, *matching*. The basic fact on which our strategy is based is the following:

¹³ Recall that μ_n converges weakly to μ if, for all $\varphi \in \mathcal{C}^0$, we have $\lim_{n \rightarrow \infty} \mu_n(\varphi) = \mu(\varphi)$.

Lemma 4.4. *Let μ_1, μ_2 two matching standard pairs, then, for each $\varphi \in \mathcal{C}^1$,¹⁴*

$$|f_*^n \mu_1(\varphi) - f_*^n \mu_2(\varphi)| \leq 4be^{ab} \|\partial_s \varphi\|_\infty \lambda^{-n}$$

Proof. It follows by a direct computation:

$$\begin{aligned} |f_*^n \mu_1(\varphi) - f_*^n \mu_2(\varphi)| &= \left| \int_b^b h(t) [\varphi(f^n(x) + \lambda^{-n} s v^s + \lambda^n t v^u) - \varphi(f^n(x) + \lambda^n t v^u)] \right| \\ &\leq 4be^{ab} \|\partial_s \varphi\|_\infty \lambda^{-n}. \end{aligned}$$

□

The above Lemma shows that the convenient topology in which to study the convergence of the pushforward of standard pairs is $(\mathcal{C}^1)'$, that is it is convenient to consider them *distributions*.

We are now ready to conclude: given two standard pairs μ_1, μ_2 , we have that $f_*^n \mu_1, f_*^n \mu_2$ are standard families in $CS_{\lambda^{-1}a}$. Note that there is some freedom in how to divide a segment of length $\lambda^n b$ in segments of length between 1 and 2. In particular one can check that, if n is large enough, one can make the division so that in one family one has a standard pair supported on $\{y + t v^u\}_{t \in [0,1]}$ and in the other one has a standard pair supported on $\{y + s v^s + t v^u\}_{t \in [0,1]}$ for some $s \in [1, 2]$. Let us choose the smallest n for which the above splitting is possible and call it n_0 ; call the two standard above mentioned standard pairs $\tilde{\mu}_{0,1}$ and $\tilde{\mu}_{0,2}$ respectively. Then we can write¹⁵

$$f_*^{n_0} \mu_1(\varphi) - f_*^{n_0} \mu_2(\varphi) = \sum_{i=1}^{m_1} \tilde{p}_{1,i} \mu_{i,1}(\varphi) - \sum_{i=1}^{m_1} \tilde{p}_{2,i} \mu_{i,2}(\varphi) + \tilde{p}_{0,1} \tilde{\mu}_{0,1}(\varphi) - \tilde{p}_{0,2} \tilde{\mu}_{0,2}(\varphi)$$

for some weights $p_{j,i}$ and standard pairs $\mu_{j,i}$. Note that $p_{j,i} \geq \lambda^{n_0} e^{-2a}$ by construction. Also

$$\tilde{\mu}_{0,1}(\varphi) = \int_{-b_0}^{b_0} h_{0,1}(t) \varphi(y + t v^u) dt; \quad \tilde{\mu}_{0,2}(\varphi) = \int_{-b_0}^{b_0} h_{0,2}(t) \varphi(y + s v^s + t v^u) dt$$

for some $b_0 \in [1, 2]$, $y \in \mathbb{T}^2$ and $h_{0,i} \in D_{\lambda^{-n_0}a}$. It follows that $h_{0,i} \geq e^{-2\lambda^{-n_0}a}$ and, for each $c > 0$ small enough,

$$\begin{aligned} \frac{h_{0,i}(t) - c}{h_{0,i}(s) - c} &\leq \frac{h_{0,i}(s) e^{\lambda^{-n_0}a|t-s|} - c}{h_{0,i}(s) - c} \leq e^{\lambda^{-n_0}a|t-s|} \frac{h_{0,i}(s) - c e^{-\lambda^{-n_0}a|t-s|}}{h_{0,i}(s) - c} \\ &\leq e^{\lambda^{-n_0}a|t-s|} \left[1 + c \frac{1 - e^{-\lambda^{-n_0}a|t-s|}}{e^{-2\lambda^{-n_0}a} - c} \right] \leq e^{\lambda^{-n_0}a|t-s|} \left[1 + c \frac{\lambda^{-n_0}a|t-s|}{e^{-2\lambda^{-n_0}a} - c} \right] \end{aligned}$$

Finally we choose c so small that

$$\gamma = \frac{c}{e^{-2\lambda^{-n_0}a} - c} \leq \lambda^{n_0} - 1.$$

so

$$\frac{h_{0,i}(t) - c}{h_{0,i}(s) - c} \leq e^{\lambda^{-n_0}a(1+\gamma)|t-s|} \leq e^{a|t-s|}.$$

¹⁴ We are using the notation $\partial_s \varphi = \langle v^s, \nabla \varphi \rangle$.

¹⁵ Note that it is not necessary that the two standard pairs have the same numbers of elements m_1 , however one can reduce to this situation by allowing some $\tilde{p}_{j,i}$ to be zero.

This means that we can write

$$\begin{aligned} \tilde{\mu}_{0,1}(\varphi) - \tilde{\mu}_{0,2}(\varphi) &= c \int_{-b_{0,1}}^{b_{0,1}} [\varphi(y + tv^u) - \varphi(y + sv^s + tv^u)] dt \\ &+ (1 - c) \left[\int_{-b_{0,1}}^{b_{0,1}} [(\tilde{h}_{0,1}(t) - c)\varphi(y + tv^u) - (\tilde{h}_{0,2}(t) - c)\varphi(y + sv^s + tv^u)] dt \right]. \end{aligned}$$

Next, let $p_* = \min\{\tilde{p}_{0,1}, \tilde{p}_0, 2\}$ and $p_{0,j} = \tilde{p}_{0,j} - p_*c$

$$\begin{aligned} p_{0,j} &= \frac{\tilde{p}_{0,j} - p_*c}{1 - p_*c}; \quad p_{i,j} = \frac{\tilde{p}_{i,j}}{1 - p_*c} \forall i > 0 \\ \mu_{0,1}(\varphi) &= \int_{-b_0}^{b_0} \frac{\tilde{p}_{0,1}h_{0,1}(t) - p_*c}{\tilde{p}_{0,1} - p_*c} \varphi(y + tv^u) dt \\ \mu_{0,2}(\varphi) &= \int_{-b_0}^{b_0} \frac{\tilde{p}_{0,2}h_{0,2}(t) - p_*c}{\tilde{p}_{0,2} - p_*c} \varphi(y + sv^s + tv^u) dt. \end{aligned}$$

Note that $\mu_{0,j}$ are standard pairs and $\sum_{i=0}^{m_1} p_{i,j} = 1$. Then, for each $n \geq n_0$, we have

$$\begin{aligned} f_*^n \mu_1(\varphi) - f_*^n \mu_2(\varphi) &= (1 - p_*c) \left[\sum_{i=0}^{m_1} p_{1,i} f_*^{n-n_0} \mu_{i,1}(\varphi) - \sum_{i=0}^{m_1} p_{2,i} f_*^{n-n_0} \mu_{i,1}(\varphi) \right] \\ &+ cp_* 4be^{ab} \|\partial_s \varphi\|_\infty \lambda^{-n+n_0} \\ &= (1 - p_*c) \sum_{i=0}^{m_1} p_{1,i} p_{2,i} [f_*^{n-n_0} \mu_{i,1}(\varphi) - f_*^{n-n_0} \mu_{2,1}(\varphi)] \\ &+ cp_* 4be^{ab} \|\partial_s \varphi\|_\infty \lambda^{-n+n_0}. \end{aligned}$$

To conclude it suffices to iterate the above formula: let $n = \ell n_0$, then

$$\begin{aligned} |f_*^n \mu_1(\varphi) - f_*^n \mu_2(\varphi)| &\leq 2(1 - p_*c)^\ell \|\varphi\|_\infty + \sum_{k=0}^{\ell-1} cp_* 4be^{ab} \|\partial_s \varphi\|_\infty (1 - p_*c)^k \lambda^{-n+kn_0} \\ &\leq C\nu^n (\|\varphi\|_\infty + \|\partial_s \varphi\|_\infty) \end{aligned}$$

for some $C > 0$ and $\nu = \max\{(1 - p_*c)^{1/n_0}, \lambda^{-1}\}$. This implies that the same estimate holds for the weak closure of CS_a , hence, for each $h \in \mathcal{C}^1$,

$$\left| \int_{\mathbb{T}^2} h(x) \varphi \circ T^n(x) dx - \int_{\mathbb{T}^2} \varphi(x) dx \right| \leq C(\|h\|_\infty + \|\partial_u h\|_\infty)(\|\varphi\|_\infty + \|\partial_s \varphi\|_\infty) \nu^n$$

which implies that the speed of mixing is exponential.

4.2. Fourier Transform. The standard pair method is very flexible and can be adapted to a large range of situations. Yet, since maps we are presently studying are linear, a much more powerful tool is available: Fourier series. Indeed, for each $k \in \mathbb{Z}^2$,

$$\begin{aligned} (\widehat{\mathcal{L}^n h})_k &= \int_{\mathbb{T}^2} e^{2\pi i \langle k, x \rangle} \mathcal{L}^n h(x) dx = \int_{\mathbb{T}^2} e^{2\pi i \langle k, A^n x \rangle} h(x) dx \\ &= \int_{\mathbb{T}^2} e^{2\pi i \langle A^n k, x \rangle} h(x) dx = \hat{h}_{A^n k}. \end{aligned}$$

Accordingly, for each $h, \varphi \in \mathcal{C}^r$,

$$\begin{aligned} \left| \int_{\mathbb{T}^2} \varphi \mathcal{L}^{2n} h - \int \varphi \right| &\leq \sum_{k \in \mathbb{Z}^2 / \{0\}} |\hat{\varphi}_k \hat{h}_{A^{2n}k}| \leq \sum_{k \in \mathbb{Z}^2 / \{0\}} \frac{\|h\|_{\mathcal{C}^r} \|\varphi\|_{\mathcal{C}^r}}{(\|A^{2n}k\| + 1)^r (\|k\| + 1)^r} \\ &\leq \sum_{k \in \mathbb{Z}^2 / \{0\}} \frac{\|h\|_{\mathcal{C}^r} \|\varphi\|_{\mathcal{C}^r}}{(\|A^n k\| + 1)^r (\|A^{-n}k\| + 1)^r}. \end{aligned}$$

For each $k \in \mathbb{R}^2$, we can write $av^u + bv^s$ (recall (4.1)). It follows that $A^n k = a\lambda^n v^u + b\lambda^{-n} v^s$ and $A^{-n}k = a\lambda^{-n} v^u + b\lambda^n v^s$. Thus

$$\|A^{-n}k\|^2 + \|A^n k\|^2 \geq (b^2 + a^2)\lambda^{2n} = \|k\|^2 \lambda^{2n}.$$

Accordingly,

$$(\|A^n k\| + 1)(\|A^{-n}k\| + 1) \geq \frac{\|k\|}{\sqrt{2}} \lambda^n.$$

We can thus conclude

$$\left| \int_{\mathbb{T}^2} \varphi \mathcal{L}^{2n} h - \int \varphi \right| \leq \sum_{k \in \mathbb{Z}^2 / \{0\}} \frac{\|h\|_{\mathcal{C}^r} \|\varphi\|_{\mathcal{C}^r} 2^{r/2}}{\|k\|^r} \lambda^{-nr} \leq C_r \|h\|_{\mathcal{C}^r} \|\varphi\|_{\mathcal{C}^r} \lambda^{-nr},$$

for some constant C_r independent on h and φ .

We have thus seen that toral automorphisms enjoy exponential decay of correlation and the speed of decay depends very much on the smoothness of the functions. In fact, with a bit more work, it is possible to obtain an even finer description of the decay of correlation. Yet, a big source of unhappiness persists: our strategy was based on the fact that the dynamics looked very simple when looked in Fourier coefficients but this happens only because the map is linear, what to do with a non linear map is very non obvious.

It would then be very desirable to obtain the above results via a different, more dynamical, strategy. In particular it would be nice if we could find a Banach space on which it is possible to study the spectrum of the operator $\mathcal{L}$ and such that the above properties can be understood as consequences of the spectral picture.

4.3. A simple class of (geometrical) Banach spaces. To construct a Banach space we first define a norm on $\mathcal{C}^\infty(\mathbb{T}^2, \mathbb{C})$ and then we obtain the Banach space by completing $\mathcal{C}^\infty(\mathbb{T}^2, \mathbb{C})$ with respect to such a norm.

Let $\partial_u \varphi = \langle v^u, \nabla \varphi \rangle$, fix $\delta > 0$, $\varphi \in \mathcal{C}_0^\infty([-\delta, \delta], \mathbb{C})$ and $h \in \mathcal{C}^\infty(\mathbb{T}^2, \mathbb{C})$ define,

$$\begin{aligned} |\varphi|_q &= \sup_{t \in \mathbb{R}} |\varphi^{(q)}(t)| \\ B_q &= \{\varphi \in \mathcal{C}_0^\infty([-\delta, \delta], \mathbb{C}) : |\varphi|_q \leq 1\} \\ \|h\|_{p,q} &= \sup_{x \in \mathbb{T}^2} \sup_{p' \leq p} \sup_{\varphi \in B_q} \int_{-\delta}^{\delta} (\partial_u^{p'} h)(x + tv^s) \cdot \varphi(t) dt. \end{aligned}$$

We will call $\mathcal{B}^{p,q}$ the closure with respect to the above norm. The first thing we want to understand is which kind of objects we obtained in the closure. The next Lemma shows that we are inside the usual space of distributions.

Lemma 4.5. *For each $p, q \in \mathbb{N}$, $p > 0$, we have $i : \mathcal{B}^{p,q} \rightarrow \mathcal{C}^q(\mathbb{T}^2, \mathbb{C})'$, where i is bounded and one-to-one.*

Proof. As usual, define $i : \mathcal{C}^\infty(\mathbb{T}^2, \mathbb{C}) \rightarrow \mathcal{C}^q(\mathbb{T}^2, \mathbb{C})'$ by $i(f)(\varphi) = \int_{\mathbb{T}^2} \varphi f$.

Let $\{\phi_i\}_{i=1}^N$ be a smooth partition of unity such that $\text{supp } \phi_i$ is contained in a ball of radius $\delta/2$ with centre x_i . Let $h \in \mathcal{C}^\infty(\mathbb{T}^2, \mathbb{C})$, for each $\varphi \in \mathcal{C}^q(\mathbb{T}^2, \mathbb{C})$ we have

$$\begin{aligned} |i(h)(\varphi)| &= \left| \int_{\mathbb{T}^2} h\varphi \right| \leq \sum_i \left| \int_{\mathbb{T}^2} h\varphi\phi_i \right| \\ &\leq \sum_i \int_{-\delta}^{\delta} ds \left| \int_{-\delta}^{\delta} dt h(x_i + sv^s + tv^u)(\varphi\phi_i)(x_i + sv^s + tv^u) \right| \\ &\leq 2\delta \|h\|_{0,q} \sum_i |\varphi\phi_i|_{\mathcal{C}^q} \leq C_{\#} \|h\|_{p,q} |\varphi|_{\mathcal{C}^q}. \end{aligned}$$

From which it follows that i is bounded and can be extended to $\mathcal{B}^{p,q}$.

Fix $g \in \mathcal{C}_0^\infty([-1, 1], \mathbb{R}_+)$, $\int g = 1$. For each $x \in \mathbb{T}^2$, $\varphi \in \mathcal{C}_0^\infty([-\delta, \delta], \mathbb{C})$ and $\varepsilon > 0$ define

$$\varphi_\varepsilon(y) = \varphi(\langle y - x, v^s \rangle) g(\langle y - x, v^u \rangle \varepsilon^{-1}) \varepsilon^{-1}.$$

Then, for $h \in \mathcal{C}^\infty(\mathbb{T}^2, \mathbb{C})$ we have

$$\begin{aligned} \int h\varphi_\varepsilon &= \int ds g(s\varepsilon) \varepsilon^{-1} \int dt h(x + sv^u + tv^s) \varphi(t) \\ &= \int dt h(x + tv^s) \varphi(t) + \mathcal{O}(\varepsilon \|h\|_{1,q}). \end{aligned}$$

Finally, suppose $i(h) = 0$ for some $h \in \mathcal{B}^{p,q}$. Let $h_n \in \mathcal{C}^\infty$ such that $h_n \rightarrow h$ in $\mathcal{B}^{p,q}$, then

$$\begin{aligned} 0 = i(h)(\varphi_\varepsilon) &= \lim_{n \rightarrow \infty} \int h_n \varphi_\varepsilon \\ &= \lim_{n \rightarrow \infty} \int dt h_n(x + tv^s) \varphi(t) + \mathcal{O}(\varepsilon \|h_n\|_{1,q}) \\ &= \int dt h(x + tv^s) \varphi(t) + \mathcal{O}(\varepsilon \|h\|_{1,q}). \end{aligned}$$

Taking the limit $\varepsilon \rightarrow 0$ we obtain

$$0 = \int dt h(x + tv^s) \varphi(t).$$

Also, differentiating in the unstable direction we have, for all $p' \leq p$,

$$0 = \int dt \partial_u^{p'} h(x + tv^s) \varphi(t).$$

Taking the sup on x we obtain $\|h\|_{p,q} = 0$. □

Before continuing it is convenient to make sure that the derivative acts in the natural way on the spaces $\mathcal{B}^{p,q}$.

Lemma 4.6. *For each $p, q \in \mathbb{N}$ the operator ∂_u is bounded as an operator from $\mathcal{B}^{p+1,q}$ to $\mathcal{B}^{p,q}$ and ∂_s is bounded as an operator from $\mathcal{B}^{p,q+1}$ to $\mathcal{B}^{p,q}$. Moreover, their kernels consists of the constants.*

Proof. The boundedness follows immediately from the definition of the norms (and integration by part in the case of ∂_s).

Next, for each $h \in \mathcal{C}^\infty$, $x \in \mathbb{T}^2$ and $\varphi \in \mathcal{C}_0^{q+1}([-\delta, \delta], \mathbb{C})$ let us define

$$h_\varphi(x) = \int_{-\delta}^{\delta} h(x + tv^s) \varphi(t) dt.$$

Then

$$\begin{aligned} \partial_u h_\varphi(x) &= \int_{-\delta}^{\delta} \partial_u h(x + tv^s) \varphi(t) dt \\ \partial_s h_\varphi(x) &= \int_{-\delta}^{\delta} \frac{d}{dt} h(x + tv^s) \varphi(t) dt = - \int_{-\delta}^{\delta} h(x + tv^s) \varphi'(t) dt. \end{aligned}$$

It follows that $\|\nabla h_\varphi\|_\infty \leq \|h\|_{1,q} |\varphi|_{q+1}$. Hence, for $h \in \mathcal{B}^{p,q}$ and $\varphi \in \mathcal{C}^{q+1}$ we have that h_φ is Lipschitz (it follows by density).

We can now study the equation

$$\partial_u h = 0$$

for $h \in \mathcal{B}^{p+1,q}$. We have $h_\varphi \in \mathcal{C}^0$ and $\partial_u h_\varphi = 0$. This implies $h_\varphi = \text{const}$. Accordingly, for each set $Q_{x,\delta} = \{x + sv^s + tv^u : t, s \in [-\delta, \delta]\}$ and $\varphi \in \mathcal{C}_0^\infty(Q_{x,\delta}, \mathbb{C})$,

$$\int_{\mathbb{T}^2} h_\varphi = \int_{-\delta}^{\delta} dt \int_{-\delta}^{\delta} dsh(x + tv^u + sv^s) \varphi(x + tv^u + sv^s) = \int_{-\delta}^{\delta} dt \int_{-\delta}^{\delta} dsh(x + sv^s) \varphi(x + tv^u + sv^s)$$

We can then set $\tilde{\varphi}_x(s) = \int_{-\delta}^{\delta} dt \varphi(x + tv^u + sv^s)$ and obtain

$$\begin{aligned} \int_{\mathbb{T}^2} h_\varphi &= h_{\tilde{\varphi}_x}(x) = \int_{\mathbb{T}^2} h_{\tilde{\varphi}_x}(y) dy = \int_{\mathbb{T}^2} dy \int_{-\delta}^{\delta} ds h(y + sv^s) \tilde{\varphi}_x(s) \\ &= \int_{\mathbb{T}^2} h \int_{\mathbb{T}^2} \varphi. \end{aligned}$$

This shows that $h - \int h$ is zero as a distribution, but then, by Lemma 4.5 it is zero in $\mathcal{B}^{p+1,q}$, thus the Lemma. Similar arguments holds for the study of the kernel of ∂_s . $\square$

Lemma 4.7. *For each $p, q \in \mathbb{N}$ we have that $\mathcal{B}^{p+1,q-1}$ embeds compactly in $\mathcal{B}^{p,q}$.*

Proof. Since the spaces are separable, it suffices to prove that each sequence $\{h_n\} \subset \mathcal{C}^\infty(\mathbb{T}^2, \mathbb{C})$, $\|h_n\|_{p+1,q-1} \leq 1$, admits a convergent subsequence. Using the language of Lemma 4.6 we see that, for each $\varepsilon > 0$, let $\{x_i\}_{i \in I_\varepsilon}$ be a finite ε dense set, then for each $h \in \mathcal{C}^\infty$, $\varphi \in B_{q+1}$ there exists x_i such that $\|x - x_i\| \leq \varepsilon$ and

$$|h_\varphi(x) - h_\varphi(x_i)| \leq \varepsilon \|\nabla h_\varphi\|_\infty \leq \varepsilon \|h\|_{1,q}.$$

On the other hand, if $|\varphi - \tilde{\varphi}|_q \leq \varepsilon$,

$$|h_\varphi(x_i) - h_{\tilde{\varphi}}(x_i)| \leq \varepsilon \|h\|_{0,q}.$$

Finally, since the set B_{q+1} is compact in B_q , there exists a finite set $\{\varphi_j\}_{j \in J_\varepsilon} \subset B_{q+1}$ such that, for all $\varphi \in B_{q+1}$, $\inf_j |\varphi - \varphi_j|_q \leq \varepsilon$. Accordingly,

$$\|h\|_{p,q+1} \leq \sup_{(i,j) \in I_\varepsilon \times J_\varepsilon} |h_{\varphi_j}(x_i)| + \varepsilon \|h\|_{p+1,q}.$$

We can then conclude by the usual diagonal trick: Note that, for each $\varepsilon > 0$, the set $\{(h_n)_{\varphi_j}(x_i)\}$ is bounded, thus contained in a compact set, hence it is possible to extract a subsequence $\{h_{n_k}\}$ such that each sequence $(h_{n_k})_{\varphi_j}(x_i)$ is converging. Accordingly, we can set $\varphi_m = 2^{-m}$, and construct recursively the sequences

$\{h_{n_m,k}\} \subset \{h_{n_{m-1},k}\}$, $\{h_{n_0,k}\} = \{h_k\}$ such that for each m there exists $K_m \in \mathbb{N}$ such that, for all $k, k' \geq K_m$,

$$\|h_{n_m,k} - h_{n_m,k'}\| \leq 2\varepsilon_m.$$

We can then choose the sequence $\tilde{h}_m = h_{n_m, K_m}$, it is easy to check that this is a converging subsequence. $\square$

We have thus described the Banach space, it is now time to study how the transfer operator acts on it.

Lemma 4.8 (Lasota-Yorke type inequality). *For each $h \in \mathcal{C}^\infty$ and $p, q \in \mathbb{N}$ we have*

$$\begin{aligned} \|\mathcal{L}^n h\|_{p,q} &\leq C_\# \|h\|_{p,q} \\ \|\mathcal{L}^n h\|_{p,q} &\leq C_\# \lambda^{-\min\{p,q\}n} \|h\|_{p,q} + C_\# \|h\|_{p-1,q+1}. \end{aligned}$$

Proof. Let $h \in \mathcal{C}^\infty$ and $\varphi \in \mathcal{C}_0^q([-a, a], \mathbb{C})$, then

$$\begin{aligned} \int_a^a (\mathcal{L}^n h)(x + tv^s) h \varphi(t) dt &= \int_{-a}^a h(x + t\lambda^n v^s) \varphi(t) dt \\ &= \lambda^{-n} \int_{-\lambda^n a}^{\lambda^n a} h(x + tv^s) \varphi(\lambda^{-n} t) dt. \end{aligned}$$

Next, we consider a $\mathcal{C}^\infty$ partition of unity $\{\phi_i\}$ of $\mathbb{R}$ such that the elements have support of size δ and $\|\phi_i\|_{\mathcal{C}^{q+1}} \leq C$, for some fixed $C > 0$. Clearly $[-\lambda^n a, \lambda^n a]$ intersects, at most, $4\lambda^n + 1 \leq 5\lambda^n$ such elements. Let t_i belong to the support of ϕ_i . Then

$$\begin{aligned} (4.3) \quad \left| \int_a^a (\mathcal{L}^n h)(x + tv^s) h \varphi(t) dt \right| &\leq \sum_i \lambda^{-n} \left| \int_{t_i-\delta}^{t_i+\delta} h(x + tv^s) \varphi(\lambda^{-n} t) \phi_i(t) dt \right| \\ &= \sum_i \lambda^{-n} \|h\|_{0,q} \leq 5 \|h\|_{0,q}. \end{aligned}$$

This proves the first inequality of the Lemma for $p = 0$. To treat $p > 0$ define $\varphi_i(t) = \sum_{j=0}^{q-1} \frac{\varphi^j(\lambda^{-n} t_i)}{j!} \lambda^{-nj} (t - t_i)^j$ and redo the above computation as follows

$$\begin{aligned} \int_a^a (\mathcal{L}^n h)(x + tv^s) h \varphi(t) dt &= \sum_i \lambda^{-n} \int_{t_i-\delta}^{t_i+\delta} h(x + tv^s) \varphi(\lambda^{-n} t) \phi_i(t) dt \\ &= \sum_i \lambda^{-n} \int_{t_i-\delta}^{t_i+\delta} h(x + tv^s) [\varphi(\lambda^{-n} t) - \varphi_i(t)] \phi_i(t) dt \\ &\quad + \sum_i \lambda^{-n} \int_{t_i-\delta}^{t_i+\delta} h(x + tv^s) \varphi_i(t) \phi_i(t) dt. \end{aligned}$$

To continue notice that

$$\left| \int_{t_i-\delta}^{t_i+\delta} h(x + tv^s) \varphi_i(t) \phi_i(t) dt \right| \leq C \|\varphi\|_q \|h\|_{0,q+1},$$

and

$$|\varphi(\lambda^{-n} \cdot) - \varphi(\lambda^{-n} t_i)|_q \leq C \|\varphi\|_q \lambda^{-nq}.$$

The above yields

$$\|\mathcal{L}^n h\|_{0,q} \leq C \lambda^{-nq} \|h\|_{0,q} + C \|h\|_{0,q+1}.$$

Next, notice that

$$\int_a^a \partial_u^p (\mathcal{L}^n h)(x + tv^s) h\varphi(t) dt = \lambda^{-np} \int_a^a (\mathcal{L}^n [\partial_i^p h])(x + tv^s) h\varphi(t) dt$$

which, remembering 4.3, implies

$$\|h\|_{p,q} \leq 5\lambda^{np} \|h\|_{p,q} + C \sum_{i=0}^{p-1} \lambda^{n(p-i+q)} \|\partial_u^i h\|_{0,q} + C \|h\|_{p-1,q+1}$$

which proves the Lemma. $\square$

The above, together with Lemma 4.7, allows to apply Theorem 2.1 and conclude that the essential spectrum of $\mathcal{L}$, when acting on $\mathcal{B}^{p,q}$ is bounded by λ^{-p} . To complete our alternative derivation of the results obtained by Fourier Transform we need to understand the discrete spectrum.

Lemma 4.9. *For each $p, q \in \mathbb{N}$ we have $\sigma_{\mathcal{B}^{p,q}}(\mathcal{L}) \cap \{z \in \mathbb{C} : |z| > \lambda^{-p}\} = \{1\}$.*

Proof. Suppose that $\mathcal{L}h = \mu h$, $|\mu| > \lambda^{-p}$. Then

$$\mu \partial_u h = \partial_u \mathcal{L}h = \lambda^{-1} \mathcal{L} \partial_u h.$$

Thus $\partial_u h \in \mathcal{B}^{p-1,q}$ is an eigenvector of $\mathcal{L}$ with eigenvalue $\partial_u h$. Doing it p time we have that $\partial_u^p h \in \mathcal{B}^{0,q}$ is an eigenvector with eigenvalue $\lambda^p \mu$, but $|\lambda^p \mu| > 1$ while the spectral radius of $\mathcal{L}$ is bounded by one, hence it must be $\partial_u^p h = 0$. But then we have $\partial_u^{p-1} h$ constant. Yet integrating we see that the constant is zero. Proceeding in such a way we have $h = \text{const}$, but then $\mu = 1$. $\square$

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CARLANGELLO LIVERANI, DIPARTIMENTO DI MATEMATICA, II UNIVERSITÀ DI ROMA (TOR VERGATA), VIA DELLA RICERCA SCIENTIFICA, 00133 ROMA, ITALY.

E-mail address: liverani@mat.uniroma2.it